Ivan Gurkov

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Education

Saint Petersburg State University 2018–2022
ScB with honors, Cognitive Sciences *GPA: 4.77*

Saint Petersburg State University 2022–2024
ScM with honors, Artificial Intelligence in Speech Technologies *GPA: 5.00*

Coursework

Master's: Python, Linear Algebra, Calculus, Probability & Statistics, Machine Learning, Neural Networks, Mathematical Linguistics, Speech Acoustics, Automatic Speech Recognition, Text-to-Speech Synthesis
Bachelor's: Cognitive Psychology, Statistical Analysis of Experimental Data, Psycholinguistics, Cognitive Neurophysiology, Neurolinguistics
Master's Thesis: Automatic Detection and Paraphrasing of Emotion Markers in Text and Speech

Teaching

Higher School of Economics Sep. – Dec. 2024

- Taught courses «Speech Technologies» and «Introduction to Computer Science» as a visiting lecturer at the HSE University
- Participated in the organization of a reading club on Natural Language Processing and Deep Learning papers

Publications

[1] N. Slioussar, I. Gurkov, and D. Chernova, "Some errors are more harmful than others: the role of type and frequency of orthographic errors in word processing," *Dialogue*, vol. 15, pp. 1158–1166, 2022.
[2] I. Gurkov and O. Mitrofanova, "Автоматическое распознавание и нейтрализация эмоцинальной речи (Automatic detection and neutralization of emotional speech)," *Фонетический лицей (Phonetics lyceum)*, pp. 15–22, 2024.
[3] P. Lyapina, I. Gurkov, D. Guseva, and M. Dolgushin, "Development of a corpus of interfered Russian speech with rhythm labeling," *Language, Music and Gesture: Informational Crossroads 2024*. Springer (accepted).
[4] N. Slioussar, D. Chernova, V. Magomedova, P. Maximova, and I. Gurkov, "The effect of spelling errors on reading tasks: a study on Russian," *The Mental Lexicon*, 2025.

Professional Development

Deep Learning School Sep. 2023 – Feb. 2024

Completed a certification from the Moscow Institute of Physics and Technology, strengthening my machine learning skills in computer vision, including classification, segmentation, and generation using PyTorch

Experience

SPBU Phonetics Lab Internship | *Python (librosa, numpy, scipy), Bash, FFmpeg* Sep. 2023 – Feb. 2024

- Developed an intonation-center detection and labeling algorithm
- Used FFmpeg, scipy and librosa to implement an audio/video sample synchronisation pipeline

- Supervised data labeling of the multimodal SPBU Irony Speech Corpus

Speech Drive | *Python, Praat, pyannote, llama-cpp, gRPC* Mar. 2024 – now

- Developing the algorithm for automatic detection and classification of emotion in speech
- Data labeling and aggregation of the emergency services speech corpora
- Developed and deployed a speech diarization pipeline combining acoustic and LLM toolchains
- Finetuning and testing our Text-to-Speech models and gRPC services

Skills

Languages: Russian, English

Programming Languages: Python, SQL, Racket, typst/LaTeX

Tools:

+Bash/Zsh
+PowerShell
git+GitHub
NeoVim
Docker
FFmpeg
Kaldi
Praat
Google Colab
- Regular Expressions

Libraries:

Pandas
NumPy
SciPy
Matplotlib
PyTorch
Scikit-learn
HuggingFace
TensorFlow (Keras)
- librosa
- NLTK
- SpaCy
- pymorphy